

Joaquín Rapela

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Education

University of Southern California 2003—2010
PhD in Electrical Engineering.
Advisors: Dr. Norberto M. Grzywacz (Neuroscience) and Dr. Jerry M. Mendel (Signal Processing)
Neuroscience Graduate Program (Studies in Neuroscience) 2001—2003
MS in Electrical Engineering 2000—2003
Universidad de Buenos Aires 1992—1998
“Licenciatura (equivalent to MS)” and BS in Computer Science.

Publications

Campagner D., . . . , **Rapela J.**, . . . , SWC GCNU Experimental Neuroethology Group (2025). *Aeon: an open-source platform to study the neural basis of ethological behaviours over naturalistic timescales*. [bioRxiv 2025.07.31.664513](#).

Rapela J., Westerfield M., Townsend J. (2018). *A new foreperiod effect on single-trial phase coherence. Part I: existence and relevance*. [Neural Computation 30\(9\):2348-2383](#).

Rapela J. (2018). *Traveling waves appear and disappear in unison with produced speech*. [arXiv:1806.09559 \[q-bio.NC\]](#).

Rapela J., Kostuk M, Rowat, P.F., Mullen, T., Chang E.F., Bouchard K. (2015) *Modeling neural activity at the ensemble level*. [arXiv:1505.00041 \[q-bio.NC\]](#).

Rapela J., Felsen G., Touryan J., Mendel J.M., & Grzywacz N.M. (2010). *ePPR: a new strategy for the characterization of sensory cells from input/output data*. [Network: Computation in Neural Systems. 21\(1-2\): 35-90](#).

Rapela J., Mendel J.M., & Grzywacz N.M. (2006). *Estimating nonlinear receptive fields from natural images*. [Journal of Vision 6\(4\), 441-474](#).

Shattuck D., **Rapela, J.**, Asma E., Chatziioannou A., Qi J., & Leahy R. (2002). *Internet2-based 3D PET Image Reconstruction using a PC Cluster*. [Physics in Medicine and Biology 47, 2785-2795](#).

Full publication list available on [Google Scholar](#).

Academic Experience

Gatsby Computational Neuroscience Unit, University College London June 2019—Present
Position: Research Engineer Fellow

- Develops, optimizes, and maintains open-source implementations of state-of-the-art machine learning algorithms for probabilistic inference in high-dimensional neural and behavioral time series (e.g., [svGPFA](#), [Linear Dynamical Systems](#)) utilizing Python, JAX, PyTorch, and Plotly.
- Spearheads curriculum development and delivers lectures on statistical neuroscience and neuroinformatics for PhD candidates at the Sainsbury Wellcome Centre, UCL ([Statistical Neuroscience 2025](#), [Neuroinformatics 2024](#)).
- Secured grant funding, directed the hiring process, and managed a team of research software engineers to execute the BBSRC-funded project [Machine Intelligence for Neuroscience Experimental Control](#).
- Mentored undergraduate fellows in statistical neuroscience for the [SURFiN Program](#): Aisha Qureshi (QMUL, CS, 2021–2022); Zimo Li (UCL, NPP, 2023–2024).
- Collaborates with experimental neuroscientists at the SWC on advanced statistical methods for characterizing neural populations electrophysiological recordings.
- Co-develops the data analytic pipeline for the [Aeon platform](#), enabling storage, management, visualization, and high-throughput statistical analysis of continuous (24/7) behavioral and electrophysiological mouse datasets.

- Designs and delivers training workshops on open science and software best practices for researchers at Gatsby and SWC (e.g., introduction to Python, code testing CI pipelines).
- Collaborates with members of SWC and Gatsby to enhance institutional research culture through the [Research Culture Working Group](#).

The Truccolo Lab for Computational Neuroscience, Brown University Aug 2017—Jan 2019
Position: Research Associate

- Discovered stereotypical discrete states in pre-ictal, ictal, and post-ictal periods of seizures separated by more than one hour, utilizing micro-electrode array layer 2/3 cortical recordings from patients with epilepsy.
- Extracted two-dimensional descriptors of high-dimensional electrophysiological recordings from patients with epilepsy that robustly separate functional ictal periods using t-SNE, XGBoost, and manifold learning.

Swartz Center for Computational Neuroscience, UC San Diego Nov 2010—July 2017
Position: Postdoctoral and Visiting Scholar

- Characterized spatio-temporal brain dynamics related to speech production from electrocorticographic (ECoG) neural recordings, analyzing synchronization and phase-amplitude coupling.
- Served as Principal Investigator for a project funded by the Center for Brain Activity Mapping (part of the B.R.A.I.N. initiative), evaluating computations of neural ensembles using high-resolution recordings.
- Discovered a new effect of temporal expectation on single-trial phase coherence of EEG data using a novel variational Bayes linear regression method.
- Found that switching attention between visual and auditory modalities generates transient arousal of attention across both modalities via time-frequency analysis.
- Developed a computer game controlled online by eye movements (electrooculography) to quantify the speed and accuracy of attention-orienting mechanics.

University of Southern California 2001—2010
Position: Research Assistant (Advisor: Dr. Norberto M. Grzywacz & Dr. Jerry Mendel)

- Investigated Bayesian models to evaluate the mathematical optimality of retinal codes.
- Developed the Volterra Relevant Space Technique (VRST) for structural spatial Volterra model estimation.
- Developed the extended Projection Pursuit Regression (ePPR) algorithm for structural characterization of visual cells.

University of Southern California Aug 2000—Aug 2001
Position: Research Assistant (Advisor: Dr. Richard M. Leahy)

- Co-developed a distributed computing Java-based system for 3D PET data reconstruction and performed source localization.

Teaching Experience

University College London (Course Organizer and Instructor) 2019—2025
 Instructor for *PyStarters* (2019, 2021).

Teaching Assistant for [NEURO019: Neuroinformatics](#) (2023).

Module organizer and instructor for the [Probability Module of the Gatsby Bridging Program](#) (2023).

Course organizer and instructor for [Neuroinformatics for SWC PhD Students](#) (2024).

Course organizer and instructor for [Statistical Neuroscience for SWC PhD Students](#) (2025).

Brown University 2017
 Teaching Assistant for *NEUR2110: Statistical Neuroscience*.

University of Southern California 2009
 Discussion Section Leader for *EE 364: Introduction to Probability and Statistics for EE/CS*.

Universidad de Buenos Aires 1998—1999
 Teaching Assistant for *Numerical Linear Algebra* and *Artificial Intelligence*.

Service	Journal Ad-hoc Reviewer: <i>Vision Research, Frontiers in Neuroscience, Journal of Perceptual Imaging.</i> Grant Reviewer: The Software Sustainability Institute (SSI).	
Funding	Biotechnology and Biological Sciences Research Council (BBSRC) 2023—2026 Co-Investigator, Grant BB/W019132/1, £800,000. Project: <i>Machine Intelligence for Neuroscience Experimental Control.</i> Center for Brain Activity Mapping (CBAM) 2013—2014 Principal Investigator, Grant 2013-023CBAM, \$30,000. Project: <i>Taking the next step toward understanding computations by neural ensembles with high resolution neural recordings.</i>	
Selected Courses	Mathematics: Topology, Fundamental Concepts of Analysis, Real Analysis, Functional Analysis. Engineering: Transform Theory, Random Processes, Digital Signal Processing, Optimization, Information Theory. Neuroscience: Control and Communication in the Nervous System, Neurobiology, Advanced Neurosciences I & II.	
Industry Experience	IBM Almaden Research Center (Staff Software Engineer) Jan 2000—Aug 2000 Oracle, Argentina (Software Engineer) June 1998—Sept 1998 IBM, Argentina & IBM Almaden (Research Fellow) Oct 1996—June 1998 IBM, Argentina (Fellowship Holder) April 1994—Oct 1996	
Programming	Python, R, C#, Matlab, Java, C/C++, Smalltalk, Pascal, Assembler.	